NASA/L'SPACE-2024-T24

Mission Definition Review

Understanding the character and origin of lunar pole volatiles

Ice Hunters, Team 24act

National Aeronautics and

Space Administration

NASA Lucy Student Pipeline

Accelerator and Competency Enabler

2024

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Table of Acronyms

CME	Coronal Mass Ejection
EDS	Electrodynamic Dust Shield
ISRU	In-Situ Resource Utilization
ESDMD	Exploration Systems Development Mission Directorate
NASA	National Aeronautics and Space Administration
LRO	Lunar Reconnaissance Orbiter
PSR	Permanently Shadowed Region
NIRVSS	Near-Infrared Volatiles Spectrometer System
NSS	Neutron Spectrometer System
TBD	To Be Determined
TRL	Technology Readiness Level
LRO	Lunar Reconnaissance Orbiter
CPU	Central Processing Unit
LEO	Low Earth Orbit
CIGS	Copper Indium Gallium Selenide
IMM	Metamorphic Multi-Junction
RTG	Radioisotope Thermoelectric Generator
ESDMD	Exploration Systems Development Mission Directorate

COTS	Commercial-Off-The-Shelf
RFA	Request for Action
ADV	Advisory
FMEA	Failure Mode and Effect Analysis
CER	Cost Estimating Relationships
NICM	NASA Instrument Cost Model
MCCET	Mission Concept Cost Estimating Tool
ERE	Employee Related Expenses
FY	Fiscal Year
FTE	Full Time Equivalent
GNC	Guidance, Navigation, & Control
GSA	General Services Administration
STEM	Science Technology Engineering Math

Ice Hunter's MDR Outline

1. Mission Definition Review

1.1. Mission Statement

The Ice Hunter lunar rover mission concept, supported by NASA's ESDMD and aligned with the Artemis Accords, aims to conduct essential surface measurements and mapping to advance understanding of lunar water distribution and abundance on the PSR of the lunar poles. This mission plans to operate on the lunar surface for a full Earth year, consistent with the Artemis mission framework. The mission's goal is to further the knowledge of the changes in volatile materials and the understanding of the compositional state and distribution of the volatile material it encounters. The mission objectives to achieve this goal are to collect 100g of ice samples from the chosen location, determine the water ice abundance in or near a PSR within the top one meter of regolith, and identify areas of higher ice concentrations with an Artemis III landing site. Furthermore, the chosen location, Haworth, was selected for its PSR in a crater showing promise of ice in previous reconnaissance missions, as well as its low slope and intermediate sunlight, allowing for exploration of its PSR. The two scientific instruments, NSS and NIRVSS, will enable the rover to achieve the mission objectives through the instrument's ability to detect hydrogen within the PSR and then drill for samples within the PSR that the instruments can more finely analyze, allowing ice samples to be analyzed.

1.2. Science Traceability Matrix

The following table will provide the link between the science goals driving the mission and the requirements that the client will set. These two contributing factors will start to root the instruments that will be used to conduct the science experiments that will be performed during the life of the mission. The instrumentation selected will allow the completion of the science goals by allowing for the detection and analysis of samples from the PSR found on the lunar poles. The goal set is centered around understanding the composition of the PSR. This will allow for a plan defined by objectives, including collecting samples, checking if the samples contain H2O, and finding a specific area for which the rover can achieve these goals. The rover will be in the Hawthorne region of the south pole and will be using spectroscopy to determine Hydrogen concentration. For example, 400-680 nm wavelengths can be detected to find areas of high hydrogen content. From this, scientists can plot regions of high ice concentration. Finding these regions with greater hydrogen content narrows down the sample collection sites for future manned missions. In line with the Artemis III mission goal to understand the transport, retention, alteration, and loss processes that operate on volatile materials, the rover will detect differences in hydrogen concentration for ground samples versus collected samples. Without disturbing the surface, a spectroscopy reading of a 3 cm^2 area will be analyzed. Then, the same area will be drilled by the rover, and another spectroscopy reading will be taken of a 100g sample to determine if a sample collection process caused a significant change in hydrogen concentration. If a significant change is detected, there should be additional research on collecting volatile ice on the moon during manned missions, especially if this ice will be used for vital operations such as oxygen supply and power generation. Certain specifications and parameters will set the standards for the science to be performed and are specified in the table below.

Figure 1. Science Traceability Matrix

Science Goals	Science Objectives	Science Measurement Requirements		Instrument Performance Requirements		Predicted Instrument Performance	Instrument	Mission Requirements
		Physical Parameters	Observables					
Understanding the character and origin of lunar poles (Primary) Understand the transport, retention, alteration, and loss processes that operate on volatile materials near and within permanently shadowed regions (NASA, "Artemis III Science.") (Secondary) Determine the compositional state (elemental, isotopic, mineralogic) and compositional	Collect 100g of ice samples at one of the 13 Artemis III landing site locations and determine the composition of the sample	Identify the concentration of Hydrogen wavelengths in a collected sample of lunar regolith	Detect a difference in water concentration from collected samples versus ground samples to predict loss of volatile material using the same 3 cm^2 area of sample collection,	Wavelength range:	250-600 nm	230-615 nm	Neutron Spectrometer System (NSS)	Vehicle shall traverse to PSR to scan/drill science locations
				Integration time	1 s	.5 s		
				Sensitivity:	10-2 to below 10-4 w/w	10-2 to below 10-4 w/w		
				Spatial resolution:	<100 µm	<100 µm		Mission must collect a minimum sample of 100 g of lunar regolith from a depth up to 1 m
	Determine the water ice abundance of at least one of the 13 Artemis III landing site locations near or within a PSR within the top 1 meter of regolith	Determine the percent concentration of Hydrogen within the top 1 meter of regolith	Detect Hydrogen wavelengths in the range of 400-680 nm	Wavelength range:	250-600 nm	200-610 nm	Near Infrared Volatiles Spectrometer System (NIRVSS)	
				Integration time:	1 s	1.2 s		
				Sensitivity:	10-2 to below 10-4 w/w	10-1 to below 10-5 w/w		Vehicle will land within 10 km of PSR
				Spatial resolution:	30 µm	15 µm		

distribution (lateral and with depth) of the volatile component in lunar polar regions (NASA, “Artemis III Science.”)	Identify areas of higher ice concentration within an Artemis III landing site	Compare concentrations of hydrogen on a 100 x 100 grid of 1 m^2 squares and identify 3 squares with the highest concentration of hydrogen	Detect areas of high ice concentration to predict ice collection sites for future missions	Wavelength range:	400-700 nm	350-750 nm	NSS & NIRVSS	
				Spatial resolution:	20 nm	10 nm		Mission shall scan lunar surface using the two science instruments
				Integration time:	1 s	1 s		
				Sensitivity:	10-2 to 10-4 w/w	10-2 to 10-4 w/w		

1.3. Mission Requirements

As a secondary payload for a discovery class mission, the constraints given by the customer for the Ice Hunter mission are primarily cost, mass and volume. The cost allotted for the Ice Hunter mission shall not exceed $225 million, which will include materials, science instruments, personnel costs and all related services, but will not include launch of cruise costs. The mass allotted for the Ice Hunter mission shall not exceed 85kg, including vehicle, science instruments, and all systems and subsystems required to accomplish the stated science goals. Finally, the total volume for the Ice Hunter mission shall not exceed a volume Length x Width x Height of 1.5m x 1.5m x 1.5m, though this is not fixed volume measurements, as the vehicle is permitted to expand and exceed these volume constraints post-deployment. Top level requirements for the mission are specified in the table below:

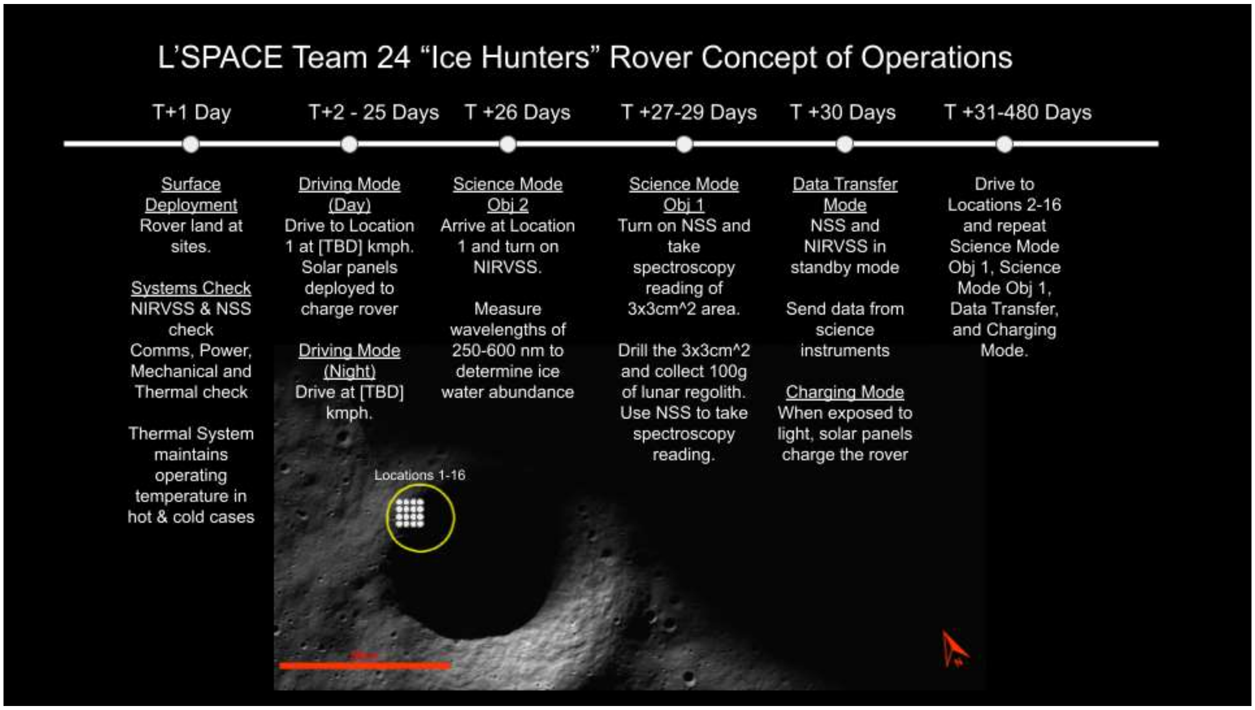
Figure 2. Mission Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
0.1	The system shall survive the lunar environment for 2 years.	In order to achieve science objectives the system needs to operate/survive	Client	SYS-03	Demonstration	All	Met
0.2	Shall investigate the abundance of near-surface water ice in PSR	Foundational science goal	Client	SYS-01.SYS-03,SYS-05,SYS-06,SYS-07,SYS-09	Test	Payload	Met
SYS-01	The system shall traverse the lunar surface by means of a rover	The rover must be able to travel distances to conduct science objectives	0.1,0.2	SYS-07,SYS-09	Demonstration	Mechanical, Electrical, GNC, Computer hardware	Met
SYS-02	The system shall not exceed a mass of 85 kg	Defined by launch provider	Launch provider	All	Inspection	All	Met
SYS-03	The system shall be able to operate at temperatures ranging from 90.15 K to 379.15 K	The rover must function in extreme temperatures of the lunar pole	0.1	SYS-07	Demonstration	Electrical, Thermal, Computer hardware	Met
SYS-04	The system shall not exceed a volume of L 1.5m x W 1.5m x H	Defined by launch provider	Launch provider	All	Inspection	Structural	Met

	1.5m						
SYS-05	The system shall collect and store data from the two science instruments on the rover	The data collected will contribute to achieving the science goal	0.1,0.2	SYS-01,SYS-03,SYS-07	Demonstration	Science, Electrical	Met
SYS-06	The system shall communicate with the LRO	After data has been collected and analyzed it will be relayed to earth	0.1,0.2	SYS-07	Demonstration	Electrical, Computer hardware, GNC	Met
SYS-07	The system shall have sufficient power to survive/operate	The rover requires power to perform and communicate all objectives	0.1,0.2	SYS-03	Demonstration	Electrical	Met
SYS-08	The system shall not exceed a cost of $225M	Defined by contracting client	Contract provider	All	Verification	All	Met
SYS-09	The system shall contain the necessary instruments to support the collection from the scientific instruments	The rover needs the ability to run experiments on the Moon’s surface	0.2	SYS-05,SYS-07	Demonstration	Science, Electrical, Computer hardware, Thermal	Met
SYS-10	The system shall not contain a Radioisotope Thermoelectric Generator(RTG)	The rover cannot have any derivative of this concept	Contract provider	All	Verification	All	Met
SYS-11	The system shall be ready for launch by September 1st, 2028	In order to align with the launch provider’s secondary payload	Contract provider	All	Verification	All	Met
SYS-12	The system shall not contain radioactive material exceeding a mass of 5g	Limited radioactive materials	Contract provider	All	Verification	All	Met

1.4. Concept of Operations

Figure 3. Concept of Operations

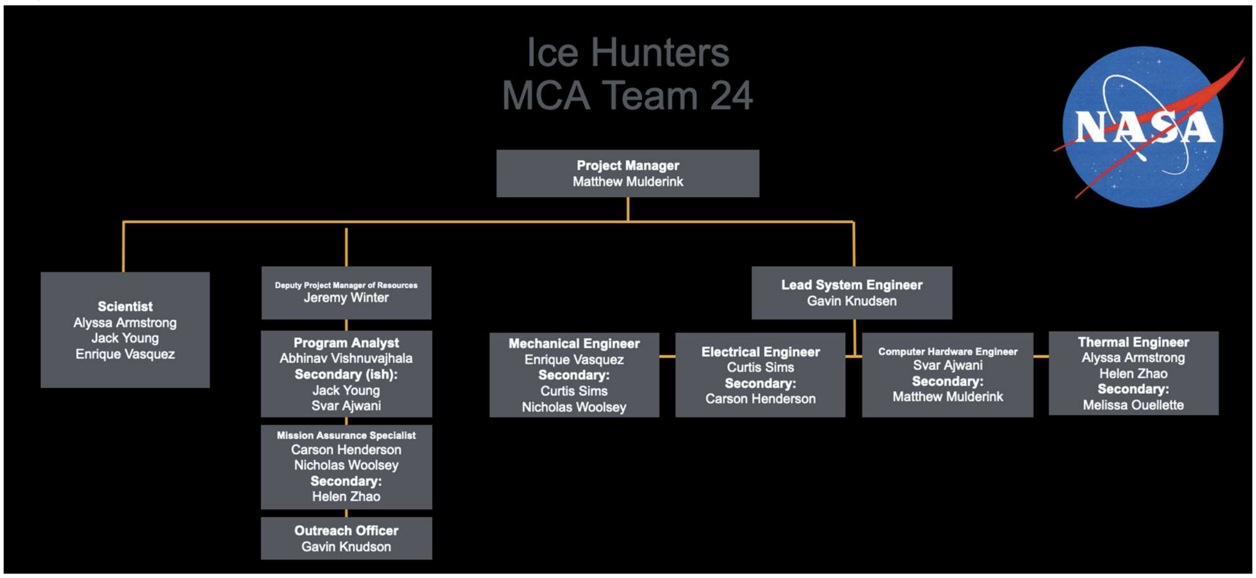


The Concept of Operations outlines the rover operation timeline from landing on the Moon to completion of data collection and transfer. At T+1 Day, the Rover will do a total systems check, ensuring science instruments, power, thermal, and communications work. A communication signal will be sent with "subsystem" = "ONLINE" for each subsystem in operation. Throughout the duration of the mission, the thermal system will maintain safe operating temperatures for both hot and cold cases. At T+2-25 Days, the rover will drive to Location 1. Over the duration of the mission, the rover will travel to 16 Science Locations, as shown by the 4x4 dot grid. These locations are a mix of near and in the PSR, to collect enough data for comparison. In Driving Mode (Day), the rover will not be in shadow and be exposed to light. The solar panels will be deployed to charge the batteries and the rover will travel at TBD kmph. At night, the rover will travel at a slower speed of TBD kmph to conserve battery and the solar panels will not be deployed. At T+ 26 days, the rover will begin data collection. NIRVSS will be turned on from standby mode and measure wavelengths of 250-600 nm and will be put in standby after data collection. This will be used to determine ice water abundance for Science Mode Objective Two. At T+27-29 Days, the rover will switch to Science Mode Objective One. The NSS will be turned on and take a spectroscopy reading of 250-600 nm wavelengths of a 3x3cm^2 area. The same area will be drilled and 100g will be collected as a sample. This NSS will take a multispectral image of 250-600 nm wavelengths of the sample. At T+30 Days, the rover will go into data transfer mode.

Data from that location will be sent to scientists for analysis. If the rover is exposed to sunlight, the solar panels will deploy and charge the batteries at this time. At T+31-480 days, the cycle will repeat as the rover collects data at all 16 locations. After data from all locations are sent to Earth, scientists can complete science Objective Three by comparing concentrations of hydrogen on a 40 x 40 km^2 grid and identify the 3 locations with the highest concentration of ice. At this stage, the mission is complete.

1.5. MCA Team Management Overview

Figure 4. Ice Hunters Team Structure



The Ice Hunter's team was changed at the end of the SRR deliverable in order to better equip its members for a specific task. Some members had up to five roles before, which was too many to focus on a specific task. Members were struggling to keep up with their different subteams and learn about the requirements needed for each subteam. Furthermore, subteam leads were struggling with the dilemma of knowing that some of the members had other commitments to the team but also did not want to completely cut out a member in fear of them falling behind or feeling excluded from the subteam. In order to try to fix this problem, members have one to two roles and a secondary role. The success of this change will not be completely known until the end of the MDR deliverable, since it is the first deliverable with this change. The general team hypothesis is that this will help most members get their work done quicker and with less hassle, though there is a worry that a struggling team member will go unnoticed and therefore will not be noticed until it is too close to the deliverable. Upon completion of the MDR, members will be asked their opinions on whether they think positively of the change or want further changes.

Furthermore, there was a change in commitment from some members of the team due to events outside of their control. This meant that some roles had to be completely swapped. Part of the reason the number of roles per person was changed was so that members could change their roles in order to cover for the change in commitments. The hardest part of these changes is that some members are still trying to learn their roles halfway through the L'SPACE program. Members who were chosen to cover these roles are learning while performing the job, which takes time away from doing the deliverables. In order to help bridge the gap between the work that needs to get done and the time available, certain team members are volunteering to help other subteams due to the smaller role that their subteam will play in the deliverable and due to a sense of generosity.

These changes will mean that members will be working more independently than before. This is based on the trust members have in one another. Members have limited time throughout the week to commit to the project, and the trust will mean that instead of spending more time double-checking others' work and trying to work a few different parts, members will instead be able to commit more of their time becoming experts in their specific field. Members' work will still be reviewed before completion, with the difference being that fewer people should have a better understanding of the work they are reviewing.

1.6. Project Management Approach

For all phases, management will include a program manager, deputy program manager of resources, chief scientist, and lead systems engineer. These will perform administrative tasks over their assigned purview over the duration of the mission. Similarly, each subteam will have a team lead assigned to overview all tasks assigned to that team. Each team will be allocated their own budget and purview, but direction will be given for all subteams by the program manager, deputy program manager of resources, chief scientist, and lead systems engineer.

Phase C

The Command and Data Handling (CDH) team will comprise 6 engineers responsible for designing and implementing the crucial systems that manage data onboard the spacecraft, supported by 4 technicians who assist in hardware integration and testing, along with 2 scientists who provide scientific oversight to ensure data integrity and processing accuracy.

Simultaneously, the Science/Payload team, pivotal for the mission's scientific objectives, includes 4 engineers focusing on the integration and optimization of payload systems, 2 technicians responsible for hands-on assembly and testing, and 5 scientists who bring specialized knowledge in fields such as astrogeology and planetary science, contributing to the mission's scientific discoveries and data analysis capabilities.

The Mechanical team, which will focus on the spacecraft's structural and mechanical integrity, comprises 4 engineers who design and validate mechanical systems, 4 technicians who execute fabrication and assembly tasks with precision, and 2 scientists who ensure that mechanical designs meet stringent performance criteria and environmental conditions.

The thermal subteam will consist of 2 design engineers responsible for thermal system design and analysis, 2 assembly, integration & test engineers who oversee the integration of thermal hardware, 3 technicians who will conduct meticulous testing and calibration, and 1 scientist who provides expertise in thermal management and heat dissipation strategies.

Lastly for phase C, the power team will include 2 engineers specializing in power system design and optimization, supported by 2 technicians who handle power system integration and testing, ensuring the spacecraft's energy needs are met under varying operational conditions.

Thus, 47 people: 18 engineers, 15 technicians, and 10 scientists are needed for phase C of the mission.

Phase D

The CDH team will continue with 6 engineers overseeing the integration and testing of data handling systems. They are supported by 4 technicians who ensure seamless assembly and integration of CDH hardware components. Additionally, 2 scientists provide oversight to validate data processing algorithms and maintain scientific integrity.

Within the Science/Payload team, 4 engineers focus on the final assembly and ensuring scientific instruments meet performance standards for space operations. They collaborate closely with 2 technicians who facilitate testing. Furthermore, 5 scientists bring specialized expertise crucial for ensuring scientific objectives are met.

The Mechanical team will consist of 4 engineers who oversee the assembly and integration of the spacecraft's mechanical systems and structures, ensuring robustness and reliability in harsh space environments. They are supported by 4 technicians in assembly tasks and conducting thorough testing. Additionally, 2 scientists provide technical oversight.

In the Thermal subsystem, 2 engineers finalize the integration and testing of thermal control systems. They collaborate closely with 2 technicians who oversee the installation and verification of thermal hardware. Additionally, 1 scientist specializing in thermal management strategies ensures optimal thermal performance across all spacecraft components.

For the Power subsystem, 2 engineers oversee the integration and testing of power systems critical for sustaining spacecraft operations alongside 2 technicians who handle the installation and testing of power hardware components.

This means that a total of 40 people are needed for phase D: 18 engineers, 12 technicians, and 10 scientists.

Phase E

The CDH team transitions to an operational phase with 2 engineers who oversee real-time data management and system operations. They are supported by 1 technician who handles routine maintenance and troubleshooting of CDH hardware. Additionally, 2 scientists continue to validate data integrity and contribute to ongoing data analysis.

In the science/payload team, 3 engineers focus on optimizing payload operations and ensuring scientific instrument functionality assisted by 1 technician who supports instrument calibration and maintenance. Furthermore, 6 scientists derive scientific insights from the collected data and guide research efforts.

The mechanical team includes 3 engineers who oversee spacecraft systems' mechanical operations and maintenance. 1 technician is required to ensure continuous functionality. Additionally, 2 scientists provide technical oversight to maintain mechanical systems' performance and reliability in lunar conditions.

Within the thermal team, 2 engineers manage thermal control system operations and monitor temperature regulation on board the spacecraft. They are supported by 1 scientist who specializes in thermal management strategies.

For the Power subsystem, 2 engineers oversee power system operations and monitor energy consumption and distribution. They are assisted by 1 technician who handles routine checks and maintenance of power systems.

Thus 25 people are needed for phase E: 12 engineers, 4 technicians, and 9 scientists.

Phase F

The CDH team in Phase F consists of 1 engineer who oversees final data management tasks and is supported by 1 technician who assists in data retrieval and system shutdown procedures. Additionally, 1 scientist provides final validation of mission data.

For the Science/Payload team, 1 engineer alongside 1 technician focuses on finalizing payload operations and preparing instruments for shutdown or decommissioning. Furthermore, 1 scientist contributes to the final analysis and documentation of mission data and scientific outcomes.

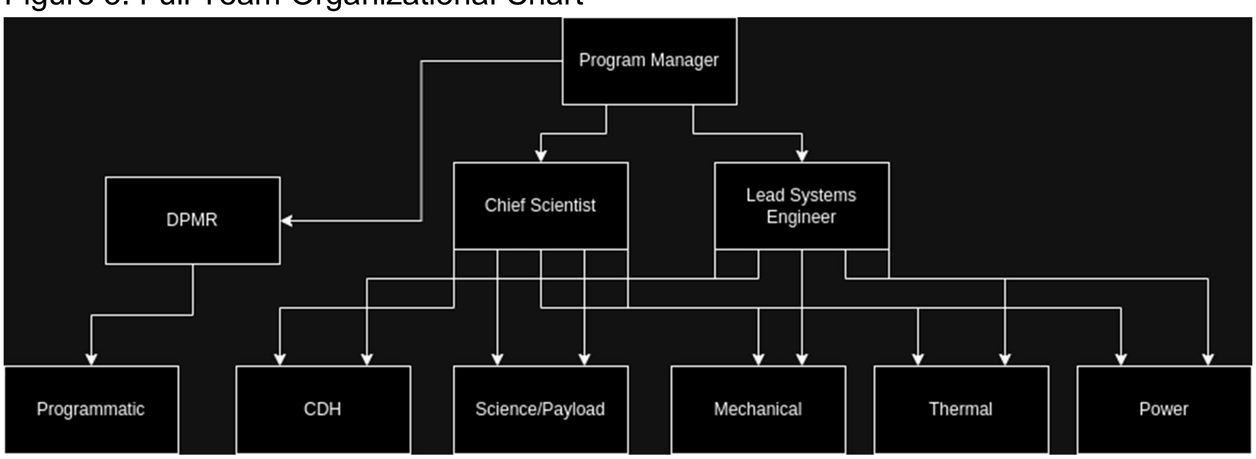
In the Mechanical team, 1 engineer, assisted by 1 technician, oversees the final mechanical operations and ensures systems are prepared for decommissioning. Additionally, 1 scientist provides technical oversight for final mechanical assessments and documentation.

The Thermal subsystem includes 1 engineer who manages final thermal control system operations and ensures systems are safely shut down. They work with 1 scientist who verifies final thermal performance data and contributes to closing reports.

For the Power subsystem, 1 engineer, supported by 1 technician, oversees final power system operations and ensures systems are safely powered down. Additionally, 1 scientist provides assistance in power system performance documentation.

This means 12 people are required for phase F: 5 engineers, 3 technicians, and 4 scientists.

Figure 5. Full Team Organizational Chart



1.7. Manufacturing and Procurement Plans

Mechanical

The mechanical subsystem will primarily consist of the mobility system, component housing/science payload deck, energy collecting, and navigating.

The mobility system of the subsystem begins with the wheels because the wheels used on a vehicle depend on the environment, there aren't many COTS options available, so this is one of the components where the team can use NASA resources to manufacture this part of the system. All six wheels would be machined from a single block of aluminum at a designated NASA facility. These wheels will be powered using a COTS product, 4 EC32 brushless motors[1] provided by Maxon, which is a reputable company in business for over 60 years. This company has also used similar motors that have been successful with spacecrafts on the surface of Mars. The last component of the mobility system is the suspension, the suspension type for the system is a rocker-bogie style, this will not yield a COTS product as the dimensions of the spacecraft will be unique in size, therefore NASA facilities will be used to develop and manufacture the suspension component.

The component housing/science payload deck will allow instruments to be mounted while maintaining housing temperatures for all other subsystems at normal operating conditions. Given the unique size of the vehicle due to client constraints, this component will be manufactured in-house using the material Aluminum 6061, a proven and widely used material in the aerospace industry due to its desirable properties. This component must have sufficient volume to store all the necessary subsystems and can be fabricated and calibrated at a designated NASA facility.

To power the different subsystems an energy collecting system is required to collect energy, this will be done using COTS parts, an EC32 brushless motor from Maxxon, which can be used to deploy a solar array along with a LA12 actuator[2] from Linak, and two coreless servo motors from BETU[3]. Together these components can raise and aim solar arrays in a given direction for the most effective energy capture given different positions and locations on the lunar surface. Linak is a family-owned business and leader in the industry for over 100 years providing exceptional products.

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[1] Maxon, "High precise drive system."

[2] Segments, "Compact and robust electrical linear actuator."

[3] Betu, "Coreless servo motor."

Another component that will contribute to achieving the science tasks and goal, this would be a drill to dig at depths up to 1 m. The device to be used will be the TRIDENT drill[4]This is a COTS product that will be contracted through Honeybee Robotics, providing innovative products since 1983. This particular drill has been designed specifically for this mission type, with plans to go on future trips to the lunar surface.

The last component to consider for the mechanical subsystem is navigating, this will require on-site imaging of the landscape. This can be accomplished by multiple cameras for navigating and/or science location reconnaissance. The main cameras will be mounted on a Leo Rover Universal Camera Mast style[5]. The two mastcams will be with The Imaging Source Color Camera[6] allowing imaging or videos from Teleskop service, this company has established a positive reputation in imaging devices for different applications. These will allow the safe navigation of the spacecraft through the PSR environment.

Power Subsystem

The Power Subsystem as specified by the system requirement PS-03 must supply the craft with power for the duration of the mission. To do this, the Power Subsystem will use several COTS parts to fulfill its role in supplying power to the rover. These parts and components are; Solar cells in array, Batteries in array, Solar Charge Controller, a Converter, and finally a Power Supply Unit. In this section, the power subsystem team will provide justification for why each part was chosen initially and also review in the same manner, other options that the Electrical Engineering team is considering to replace some parts and raise the TRL of the system.

Firstly, the solar cells in the array that were chosen are the ZTJ triple junction solar cell from Rocket Labs. These solar cells were chosen from this company because of Rocket Labs' proven reliability in the satellite market with over 1100 installations powered by their technology in orbit as well as product testing through a thermal cycle before delivery. On top of this, the Electrical Engineering team decided to go with a COTS triple junction solar cell as the development of an in-house triple-junction cell would have taken more time and money than using a COTS component. The lead time for this component is around 18 months and is the longest expected lead time. In case this component cannot be used, The Sparkwing solar panel from Sparkwing Space will be used in its stead. This company was chosen as a backup because it has over 40 years

[4] Honeybee Robotics, “Drill products.”
[5] Leo Rover, “Universal camera mast.”
[6] Teleskop-express, “The Imaging Source.”

of history supplying satellite solar panels and its product has flown on over 85 missions while also maintaining a similar amount of power supply for the craft.

Moving on, the batteries that are currently specified in the Power Subsystem are the Eaglepicher 55 Ah lithium ion battery cells. These batteries were chosen because of Eaglepicher's legacy as a satellite battery manufacturer with over 3 billion installed cell hours in space and because they were designed with a space focused use case in mind, creating specifications that align with the craft's needs. Eaglepicher also has a history with NASA as it has provided batteries for missions like Mars Pathfinder and Mars Odyssey further increasing its credibility (Eaglepicher, 2024). The lead time for this product will be around 5 months as industry standard for a custom or low volume battery. The backup component that will be used if this component cannot be delivered is the Optimus-80 from AAC Clyde Space. This company has a history of providing the UK Space Agency with components and has supplied components for, and built several satellites with their components on board that are currently operating in orbit like the YMIR-1 (AAC Clyde Space, 2024).

The next part in the system is the solar controller. For this, the Electrical Engineering team chose the AIMS Power 10 AMP Solar Charge Controller. This part and company were chosen for several reasons initially but will likely be changed in place of a more space dedicated component. This component initially fits the thermal, environmental and power needs of the system, however, this solar power converter was not designed by a company in the aerospace business and while reputable in the solar business is not yet proven in space. Therefore, in the future this component may be replaced with an all in one power supply and solar controller like the STARBUCK-MINI from AAC Clyde Space which currently has several operational cubesats in orbit and has also worked closely with the UK space agency. As this part replaces the next two components as well, the backup part supplier and lead time for this component will be discussed at the end of the justification section.

The next component that the Electrical Engineering team chose for the Power Subsystem was the DC to AC inverter. The EE team decided on the 24V to 120V inverter made by and sourced from inverter.com. This fulfills several requirements such as the power conversion and input/output voltage requirement, but is lacking in several aspects such as TRL. On top of this, the company's lack of experience making products for the space industry, the products unproven reliability at low temperatures and the reputability of the company decrease the confidence the Electrical Engineering team has in this component. For these reasons this component will also likely be replaced by the STARBUCK-MINI from AAC Clyde Space as it fulfills the requirements and is produced by a more reputable and proven company.

Finally, the power supply that the system is using to deliver power is the MeanWell SPV-300 series AC/DC Power supply. This power supply fits all of the power requirements of the system but the selected company does not meet NASA expectations as it has not constructed any components for space and therefore there are several points where this product could fail such as vibration from launch, radiation exposure, and temperature resistance which would need to be tested. To remedy the extended certification process for this part the EE team will likely also switch this power supply in lieu of the STARBUCK-MINI from AAC Clyde Space as this company is reputable in the aerospace industry. This component has a lead time of 9-14 months however if this component cannot be sourced, there are two backup parts from different suppliers, the first is the SuperNova PCDU from Bradford Space which makes an almost identical product but may force a redesign of the battery pack as it only takes in power at 28 volts from this source. This company has a similar background to AAC Clyde Space and has experience similar to that of it as well. The second option for the previous three parts is to use the previously selected options. This is not a good option in terms of TRL and reliability but provides a nearly guaranteed delivery as they are all widely available commercial parts.

CDH Subsystem

The CDH subsystem will use the following components: Single Board Computer, Data Bus, Transmitter, Receiver, Storage, and Software.

The single-board computer chosen is the BAE Systems RAD5545 Radiation-Hardened Processor. The supplier will be L3Harris Technologies. L3Harris Technologies is a historically reliable supplier for spage-grade CPUs, as NASA's perseverance rover is currently operating using a BAE CPU supplied by them. If procurement of the RAD5545 is not possible, the team's backup will be a RAD750 which is still compatible with all of the other components. An exact estimate of manufacturing time was not found, but according to multiple sources, radiation-hardened CPUs take a significant amount of time to be manufactured. In this case, it seems as if the RAD5545 will take around 3 months to procure. This is a safe estimation based on quote time, international shipping and potential delays.

The data bus chosen is SpaceWire's data buses. The primary supplier will be STAR-Dundee. STAR-Dundee seems the most reliable out of the SpaceWire suppliers researched, and they possessed a handful of documentation regarding SpaceWire. The backup supplier is Cobham Advanced Electronic Solutions. Standard products' lead times are estimated to be around 4-12 weeks. Thus, 12 weeks is the baseline estimate for procuring a SpaceWire data bus.

The antenna/(transmitter + receiver) will be the BAE Systems Ka-band multi-orbit LEO optimized antenna. The supplier for the antenna will be ViaSat. The backup supplier will be L3Harris Technologies. BAE systems have proven to be trustworthy and create well-performing radiation-hardened equipment. The estimated lead time when compared to similar antennas from different vendors is approximately 3 months.

The storage chosen is 2GB flash storage supplied by Phison USA. This company manufactured the storage on the Perseverance rover[7]. The company should be able to provide it completely radiation-hardened, and the team will be responsible for testing. The lead times for this was especially difficult to find, but it should not take longer than 3 months.

The software being used is Linux OS. Linux is a commonly used operating system among different space technologies, such as the SpaceX Satellite network. NASA will need to program this software in-house. The estimated lead time for this is however long it takes the developers to code it.

Note: the lead times are fairly rough estimates. It was very difficult to find credible sources online that could give lead times for even other models of each component. However, since they are all COTS, three months should be a generous estimate for shipping and delivery.

Thermal Subsystem

The thermal system will be using the following components: heaters, radiators, a thermal switch, and a protective thermal coating. The heater used is an Omega KHLVA, PLM-Series. This is a flexible heater that comes in a variety of sizes. The chosen model is PLM-112/2-P heater which is 1 inch in width and 12 inches in length. The order time is 1 week shipped from India to the United States. The team chose Omega as the supplier because the website offers detailed information about the heater's shipping time, cost, and size. The shipping time can range from 1 week to 11 weeks, so the team chose a heater with the shortest shipping time.

The radiator chosen was the Q-Rad Deployable Radiator from redwire space, based in Jacksonville, Florida. The company has great heritage, with involvement in missions such as Artemis I and Mars 2020. This component is off the shelf, however, shipping time is unclear despite extensive research on both the company and third-party websites.

[7] Mellor, “Fresh Details on the Flash That Sits inside Mars Perseverance Rover.”

The Sierra Nevada Thin Plate Heat Switch passively controls the thermal system. It is situated between the electronics and radiator and increases conductance when electronics become too hot to dissipate heat into the radiator. Sierra Nevada Space Technologies was used because its components were low-cost and self-regulating. There are locations all across the US, which will decrease shipping time. The website does not have detailed information on manufacturing and shipping time, however, it can be estimated to take about a month. The components are manufactured and shipping time across the US takes about a month at most.

The thermal protective coating chosen is the Socomore Aeroglaze A276 Polyurethane Coating in white. The manufacturer, Ellsworth Adhesives, ships the paint in 1-2 business weeks. The paint is already made so there is no manufacturing time. Buying the paint directly from Socomore is another option, however, the shipping times are unclear on the website.

Science

The science instruments selected for the mission's tasks include two instruments, these two instruments will be COTS instruments because of their high TRL and success on similar missions.

The first instrument selected to achieve the science goal is the Neutron Spectrometer System (NSS), this will be contracted through the company Astrobotic. This company has had a part on multiple missions including landers and rovers, because of their success this makes for an excellent product choice for the first of the two science payloads.

The second instrument of the two instrument's max payload will be the Neutron Infrared Volatiles Spectrometer System (NIRVSS), this will also be contracted through Astrobotic to develop this science instrument, which as stated above has had great success with this product on previous missions.

1.8. Risks and Safety

1.8.1. Risk Analysis

Figure 6. Risk Summary

MDR Risk Summary

ID	Summary	L	C	Trend	Approach	Risk Statement	Status
1	Mechanical	2	5	↓	M	Wheels of rover fail to endure unexpected rocky terrain	Active
2	Mechanical	3	5	↓	M	Solar arrays do not deploy	Active
3	Power	3	5	↓	M	Solar cells are kept in a PSR for too long leading to a loss of power	Active
4	Power	2	2	↓	M	Lunar dust particles settling on solar panels preventing efficient charging of the rover	Active
5	CDH	3	5	↓	M	Radiation exposure causes damage to semiconductor devices leading to corrupted data or component, instrument, or vehicle loss or damage	Active
6	CDH	2	5	↓	M	An event occurs which causes the rover to cease communication	Active
7	Thermal	1	4	↓	M	System fails to prevent the system from overheating or freezing	Active
8	Thermal	2	5	↓	M	Thermal switch fails to regulate temperature through change in lunar environment	Active
9	Instrumentation	1	5	↓	M	Thermal regulation system fails to keep the NIVRSS and NSS at appropriate operating temperatures	Active
10	Instrumentation	2	4	↓	A	The LRO fails to fly over the rover to relay signals back to Earth	Active
11	Programatics	3	3	NEW	M	A new person must be found to fill in for an administrative role leading to delays	Active
12	Programatics	3	3	NEW	M	An injury takes place during the course of the mission	Active
13	Planetary	3	3	NEW	M	The rover contaminates the lunar surface	Active

	Protection						

Figure 7. Risk Matrix

L = Likelihood (1-5)		LxC Trend		Approach		Criticality
1 = not likely		↓ **-** Decreasing (improving)		A - accept		**HIGH**
5 = extremely likely		↑ - increasing (worsening)		M- mitigate		**MED**
C = Consequence (1-5)		→ **-** unchanged		W - watch		**LOW**
1 = low consequence		**NEW** - added this month		R - research		
5 = high consequence						

LIKELIHOOD						
	5					
	4					
	3			11,12,13		3, 5
	2		4		10	1
	1					9

1	2	3	4	5
CONSEQUENCES				

(1) Given that the terrain of the Lunar surface near the south pole could be harsh near Haworth Crater, there is a possibility of the rover's wheels being damaged, resulting in unreliable equipment and lost mobility of the rover, thus ending the ability to collect data samples.

(3) Given that the rover is exploring areas within a PSR, there is a possibility that the solar cells go too long without a charge, resulting in a total loss of power, thus ending any ability to use the rover.

(3) Given that the rover will be kicking up dust from driving and operating it's drills, there is a possibility that lunar dust will accumulate on the solar panels, resulting in a less efficient power capture from the solar cells, thus decreasing the efficiency of the solar system.

(9) Given the operating temperature of the NIVRSS is 10° to 35° in the moon environment that ranges from -183° to 106°, there is a possibility that the NIVRSS will overheat or freeze over, resulting in damaged components, thus terminating the ability to analyze the composition of samples.

(10) Given the LRO has a large flight path and unexpected re-directions can occur, there is a possibility that the LRO becomes off path, resulting in no communication between the rover and Earth, thus halting the rover's ability to move forward with drilling.

1.8.2. Failure Mode and Effect Analysis (FMEA)

Figure 8. FMEA Chart

Function	Failure Mode	Effects	Sev	Cause	Occ	Prevention	Det	RPN	Actions
Power	Solar cells fail to charge batteries.	System loss of power resulting in mission failure.	10	Timing the rover's charging times incorrectly could cause the rover to be stranded in a shadowed region without the ability to recharge using solar panels.	3	Keeping track of the battery drain from collecting & sending data, driving and keeping the whole system powered. Planning driving routes with charging stops and time margin to reach a charging spot during the day.	9	270	If this failure continues, the number of science locations can decrease and increase the number of charging locations. Redundant batteries can be used, however, they carry extra mass.
CDH	Radiation exposure leading to data and equipment malfunction.	Could permanently damage some or all electronics.	8	Not testing the effectiveness of the radiation shielding could cause charged particles to disrupt the electronic system.	7	Use a test bed to ensure radiation shielding design prevents electronic damage.	4	224	Radiation is always present, so include redundant radiation shielding or a backup CPU.
	Radiation exposure leading to data malfunction.	Cause interference with data transfer, or could cause stored data loss before data transfer.	7	In the time between data collection and data transfer, radiation exposure that is damaging to electronics could also corrupt the data stored in the rover's drive.	5	Reduce the time between data collection and transfer. Add extra radiation protection around the rover's storage drive.	3	105	Back up data into a secondary drive. Add extra radiation shielding to the hard drive. Coordinate data collection timing with the satellite's data transfer range timing to reduce the time the data is stored.
Mechanical	Wheels of the rover wear over time in lunar terrain.	Science objective 3 cannot be accomplished if the rover is unable to travel to multiple locations.	8	Using a testing environment that does not accurately represent the terrain and environment of the moon. Not being able to test the wheels for 3 years.	6	Use wheel design and material with heritage, especially 3+ years on the Moon.	9	432	Reduce the number of science locations. Increase the number of wheels (most likely not possible due to mass).

1.8.3. Personnel Hazards

Hazard	**Mitigation**
Debris/parts flying into eyes	Safety glasses with side shields must be worn
Clothing, jewelry, and hair can get stuck in machinery or parts	Do not wear loose clothing, loose neckwear or exposed jewelry while operating machinery. Pull back and secure long hair
A serious injury occurs and there is nobody around to help	Do not work alone. Use the buddy system
Heavy object falls on foot	Wear strong close-toed shoes at all times
Sharp machinery, moving objects, and tools can cut and cause serious injury	Keep hands and other body parts away from moving machine parts, workpieces, and cutters that you are not using. Use tools for their designed purpose only
Machinery causes injury because safety mechanisms are not in place	If guards or safety mechanisms are present do not remove or disable them. Ensure all machinery has appropriate guards in place to prevent accidental contact. Regularly inspect guards for damage or wear
Inexperience with a machine poses a large risk	Do not operate any machine unless authorized, and do not set up or operate machinery if you are not trained and familiar with that setup
Attempts to clean and perform maintenance on machinery cause injury	Do not attempt to oil, clean, adjust, or repair any machine while it is running. Ensure the machine is in the "OFF" position. Use a brush to remove short, discontinuous types of chips--not hands, fingers, or rags
Injury due to working/moving equipment	Do not leave the machine until it has stopped running. Do not leave a machine until it has come to a complete stop
Emergency stop injury	Do not try to stop the machine with your

	hands or body. Press the emergency stop button and wait until the machine has come to a complete stop to proceed
Injury due to dull or broken tools	Check tools and machines before use to ensure they are safe to use. Always use correct speeds and feeds
Parts or tools become loose and become projectiles	Always check work and cutting tools on any machine are clamped securely
Injury due to carelessness	Concentrate on the work and do not talk unnecessarily while operating the machine. Make sure you are working with a calm and clear mind
Gloves get caught in rotating machinery and cause serious injury	Always remove gloves before turning on or operating any machine. If material is rough or sharp and gloves must be worn, place or handle material with the machine turned off. Never wear gloves or use rags to clean the workpiece or any part of a machine that is running. Rotating tools or parts can grab gloves and rags and pull you into the machine.
Cuts and injuries from chips and shavings	Never handle chips with your bare hands or fingers. Chips are extremely sharp and can easily cause cuts
Strain or injury from lifting heavy objects	Lift heavy objects with your legs and not your back. Additionally, ask for help
Hazardous walking environment caused by chips and debris	Floors, machines, and other surfaces should be kept free of wood and metal chips, dirt, and debris. Proper footwear should also be used
Flammable and hazardous materials	Flammable and combustible liquids such as oils, gasoline, cleaners, solvents, paints, and thinners should be stored in UL-approved liquid storage cabinets. Chemicals should be stored in cabinets approved for that use. Do not store incompatible chemicals together. Safety data sheets should be kept in the shop area for all chemicals used

Fire hazards from hot work areas	Hot work areas, such as welding and grinding areas, should be separated by at least 35 feet from flammable and combustible materials. Sparks should be controlled through the use of screens, if necessary
Risk due to improper signage	Emergency contact numbers are posted, hazard signs are posted and current, any additional warning signs are posted, and all containers are clearly labeled
Chemical or foreign body in eye	Emergency eyewash and shower are available and clearly labeled where chemicals are stored and used
Electrocution and electrical hazards	Ensure all electrical equipment is properly grounded. Inspect electrical cords and plugs for damage before use. Do not overload circuits. Use Ground Fault Circuit Interrupters (GFCIs) where necessary. Ensure that only qualified personnel perform electrical work
Excessive noise and hearing damage	Hearing protection is provided in high-noise areas.
Ergonomic hazards	Ergonomic workstations are provided to reduce repetitive strain injuries.
Chemical exposure	Proper ventilation systems are used to reduce inhalation hazards and personal protective equipment (PPE) is available in chemical areas
Radiation hazards	Proper shielding and dosimeters are used to monitor radiation exposure. Only trained personnel handle radioactive materials. Appropriate warning signs are posted in areas where radiation is present
Laser hazard	Appropriate eye protection is worn when operating lasers in controlled areas. Lasers have proper warning labels
Slips, trips, and falls	Walkways must be kept clear of obstacles and spills. Anti-slip mats are placed in

	areas prone to slip conditions. Proper footwear is used
Heat-related hazards and exhaustion	Adequate hydration areas and rest breaks for employees are provided, especially for those working in hot environments. Temperature and humidity levels are monitored. Use fans or air conditioning to cool work areas
Biological hazards	Proper sanitation and hygiene practices are used. Training is provided on handling biological materials. Availability of personal protective equipment such as gloves and masks is ensured
Fire and explosion hazards	Fire extinguishers are maintained and are easily accessible in all areas of the facility. Regular fire drills are conducted and employees are trained

https://tulsa.okstate.edu/helmerich-research-center/equipment/docs/hrc-machine-shop-safety-tips.pdf
https://www.emcinsurance.com/losscontrol/techsheet/machine-shop-safety-hazards-and-rules
https://www.ehs.washington.edu/system/files/resources/prvaslbchklst.pdf

1.9. Schedule

1.9.1. Schedule Basis of Estimate

The current mission schedule plans a launch date for September 1st, 2028 and an arrival date of September 5th, 2028, which exactly meets the L'Space mission constraint. However, it is likely that the team will be ready for launch well before the current given estimate, and further analogous comparison with similar lunar missions is needed for more accurate progress projection. Phases A and B are excluded from the timeline, and Phase C is assumed to directly start after the conclusion of the L'Space Program. Phases A and B have been excluded from the timeline. Phases C and D utilize data from the Psyche timeline, chosen for its similarities to the Ice Hunter mission in terms of program structure and scope. Phase E utilizes data from SpaceX and estimated timelines for the proposed rover at the Haworth landing site.

The below schedule (including KRB, SRB, and LCR specifics) are based on the NASA Project Management handbook and L'Space team resources. To reiterate, further *comparison with similar missions (such as Curiosity and VIPER) is needed to refine these estimates.*

1.9.2. Mission Schedule

The major milestones for the mission are based around the following timeline:

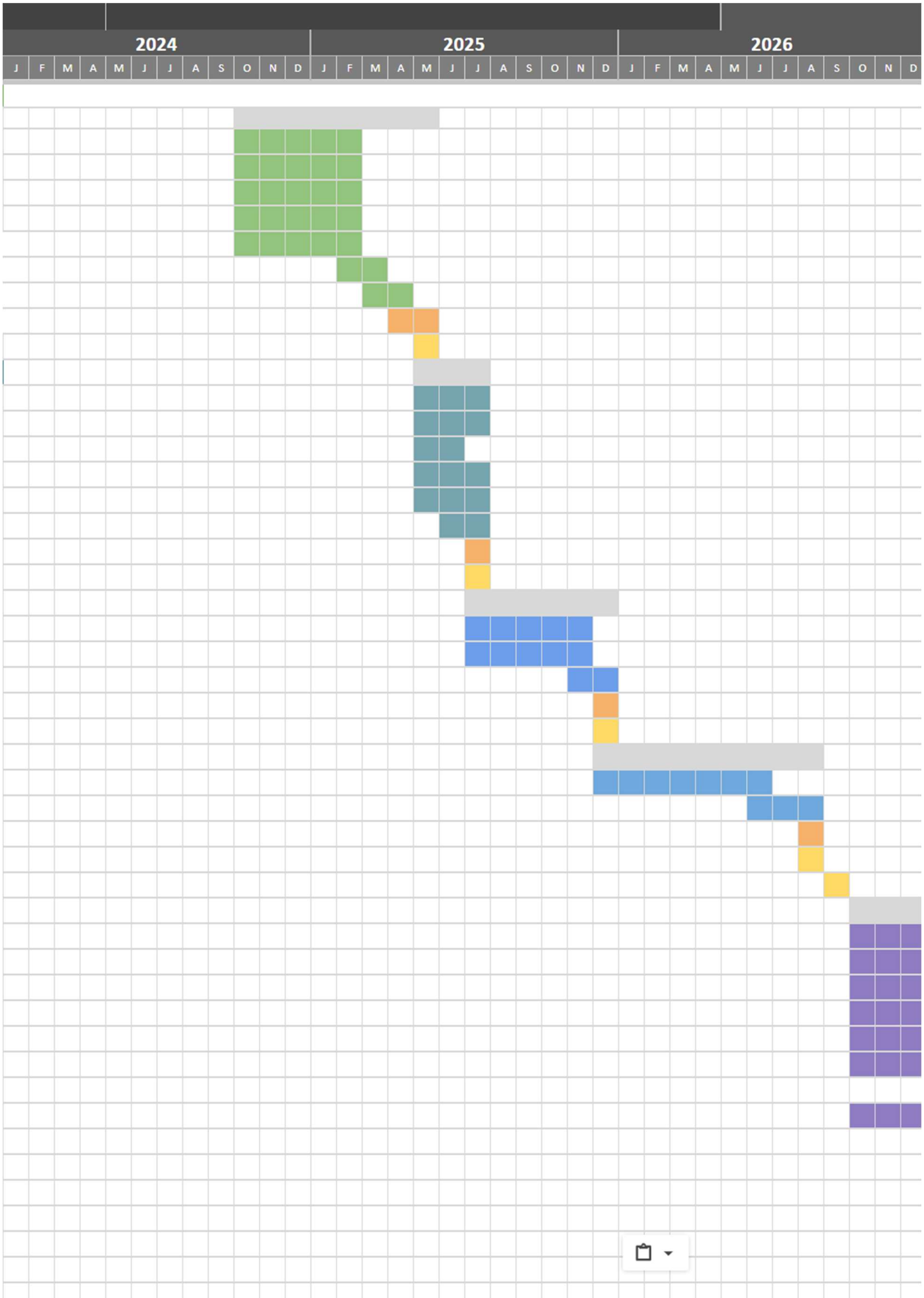
Figure 7. Table of Milestones

Milestone	**Date**
Full-scale assembly of rover instruments and subsystems complete	5/9/25
Completion of Initial Component Orders	7/16/25
Completion of Fabrication for Test	12/18/25
Systems Integration Review (SIR)	8/28/26
Critical Design Review (CDR)	9/30/26
Subsystem Testing Complete	6/11/27
Integrated Testing Complete	9/17/27
Launch Readiness Review (LRR)	12/17/27
Launch Date	9/1/28

Figure 9. Gantt Chart

1	**Final Design Post PDR**		0%	10/1/24	5/9/25	221	22
1.1	Detailed design of Mechanical System	Mechanical	Not complete	10/1/24	2/7/25	130	13
1.2	Detailed design of Electrical System	Electrical	Not complete	10/1/24	2/7/25	130	13
1.3	Detailed design of Thermal System	Thermal	Not complete	10/1/24	2/7/25	130	13
1.4	Detailed design of Computer/Hardware	Computer	Not complete	10/1/24	2/7/25	130	13
1.5	Detailed design of Science Instrument Integratic	Science/Mechanical	Not complete	10/1/24	2/7/25	130	13
1.6	Simulation Testing	All	Not complete	2/7/25	3/28/25	50	5
1.7	Completed CAD of all components	All	Not complete	3/28/25	4/16/25	20	2
1.8	Schedule Margin			4/17/25	5/8/25	22	3
1.9	◆Full-scale CAD assembly of rover subsystems		Not complete	5/9/25	5/9/25	1	
2	**Ordering and Procurement**		0%	5/12/25	7/16/25	66	14
2.1	Order Mechanical System Parts	Mechanical	Not complete	5/12/25	7/1/25	51	6
2.2	Order Electrical System Parts	Electrical	Not complete	5/12/25	7/1/25	51	6
2.3	Order Thermal System Parts	Thermal	Not complete	5/12/25	6/17/25	37	4
2.4	Order Computer Hardware System Parts	Computer	Not complete	5/12/25	7/1/25	51	6
2.5	Order NSS and NIRVSS	Science	Not complete	5/12/25	7/1/25	51	6
2.6	Verify Test Facility Availibility	Programmatics	Not complete	6/9/25	7/1/25	23	3
2.7	Schedule Margin			7/2/25	7/15/25	14	2
2.8	◆ Completion of Initial Component Orders		Not complete	7/16/25	7/16/25	1	1
3	**Manufacturing and Fabrication**		0%	7/17/25	12/19/25	156	14
3.1	Fabricating of all Subsystems	All	Not complete	7/17/25	11/14/25	121	13
3.2	Fabricating electrical testbed	Electrical	Not complete	7/17/25	11/14/25	121	13
3.3	Preliminary tests for integration	All	Not complete	11/14/25	12/5/25	22	3
3.4	Schedule Margin			12/5/25	12/18/25	14	2
3.5	◆ Completion of Fabrication for Test		Not complete	12/18/25	12/19/25	2	1
4	**Integration and Assembly**		0%	12/22/25	8/28/26	250	22
4.1	Assembly of each Subsystem	All	Not complete	12/22/25	6/30/26	191	20
4.2	Integrating subsystems into test rover	All	Not complete	6/30/26	8/5/26	37	4
4.3	Schedule Margin			8/6/26	8/27/26	22	3
4.4	◆ Systems Integration Review (SIR)		Not complete	8/28/26	8/28/26	1	1
4.5	◆ Critical Design Review (CDR)			9/30/26	9/30/26	1	1
5	**Subsystem Testing**		0%	10/1/26	6/11/27	254	28
5.1	Electrical Tests on Testbed	Electrical	Not complete	10/1/26	5/14/27	226	23
5.2	Static and Dynamic Testing	Mechanical	Not complete	10/1/26	5/14/27	226	23
5.3	TVAC testing of internal components	Thermal/Mechanical	Not complete	10/1/26	5/14/27	226	23
5.4	Drill Tests	Mechanical	Not complete	10/1/26	5/14/27	226	23
5.5	Communication Tests	Computer	Not complete	10/1/26	5/14/27	226	23
5.6	Power cycle testing	Power	Not complete	10/1/26	5/14/27	226	23
5.7	Mechanical wheel test in Moon Yard	Mechanical	Not complete	1/11/27	5/14/27	124	13
5.8	Instrument Calibration Tests	Science/Mechanical	Not complete	10/1/26	5/14/27	226	23
5.9	Schedule Margin			5/14/27	6/10/27	28	3
5.1	◆ Subsystem Testing Complete		Not complete	6/11/27	6/11/27	1	1
6	**System-level testing**		0%	6/11/27	9/17/27	99	14
6.1	TVAC testing for complete rover	Thermal	Not complete	6/11/27	7/2/27	22	3
6.2	Vibration Tests for complete rover	Mechanical	Not complete	7/2/27	7/16/27	15	2
6.3	Simulated Emergency testing	All	Not complete	7/16/27	7/30/27	15	2
6.4	Mock Mission Testing	All	Not complete	7/30/27	9/3/27	36	4
6.5	Schedule Margin			9/3/27	9/16/27	14	2
6.6	◆ Integrated Testing Complete		Not complete	9/17/27	9/17/27	1	1

7	**Launch Preparation**		0%	9/20/27	12/17/27	89	15
7.1	Create final summary of risk analysis	Programmatics	Not complete	9/20/27	10/4/27	15	2
7.2	Verify mission operational status	Systems	Not complete	9/20/27	10/25/27	36	4
7.3	Prepare Launch Readiness Review	All	Not complete	9/20/27	12/3/27	75	8
7.4	Schedule Margin			12/3/27	12/17/27	15	2
7.5	◆ Launch Readiness Review (LRR)		Not complete	12/17/27	12/17/27	1	1
7.6	◆ Launch Date		Not complete	9/1/28	9/1/28	1	1


2024
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1.10. Budget

1.10.1. Budget Overview

The Ice Hunters Budget mission has a project cost of $173,711,430. This total then has a total cost margin of $20,791,000, meaning that the real cost can total up to $194,502,430. The budget breakdowns into the following categories: total personnel cost is projected to be $25,152,201, total travel costs are $264,525, total outreach costs are $2,232,787, total direct costs are $138,835,723, and total MTDC is $ 82,561,942. In viewing this data, most of the cost lies in the direct costs, with the personnel budget accounting for a lot of the rest of the budget.

Link to budget template

Figure 10. Final Cost Breakdown

NASA L'SPACE Mission Concept Academy Budget - *Ice Hunters (Team 24)*

Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Cumulative Total
FINAL COST CALCULATIONS							
Total F&A	$ 1,104,727	$ 2,079,957	$ 2,102,993	$ 1,616,717	$ 237,824	$ 83,977	$ 7,226,194
Total Projected Cost	$ 27,864,301	$ 47,818,980	$ 48,880,205	$ 34,792,822	$ 9,242,085	$ 5,113,037	$ 173,711,430
Total Cost Margin	$ 4,210,000	$ 6,014,000	$ 6,029,000	$ 3,234,000	$ 694,000	$ 610,000	$ 20,791,000
	15.1%	12.6%	12.3%	9.3%	7.5%	11.9%	
Total Project Cost	$ 27,864,301	$ 47,818,980	$ 48,880,205	$ 34,792,822	$ 9,242,085	$ 5,113,037	$ 173,711,430

1.10.2. Budget Basis of Estimate

In researching for a mission to search for volatile deposits in permanently shaded regions of the lunar surface, several key ground rules, assumptions, and drivers were considered in the budgetary planning. The team assumed that the rate of inflation would remain constant throughout the course of the mission, based on historical inflation data and current economic forecasts, estimating a stable inflation rate of 2.6 percent per year. Personnel costs were projected to increase at this rate, based on industry standards for aerospace and scientific research fields with the hope of ensuring competitive compensation would attract and retain top talent for the mission.

Travel costs were calculated using average lowest-cost coach flights of average distances and frequencies for mission support with the fewest employees necessary, with estimates derived from the General Services Administration (GSA) travel rates and historical data on airfare and lodging costs for per diem. Outreach activities were assumed to be conducted primarily through digital platforms and local events to minimize costs, with historical outreach program costs and the increasing trend towards virtual engagement informing these estimates.

Direct costs associated with materials, equipment, testing, and facilities were assumed to remain stable with minor adjustments over the course of the development periods based on market analysis and publicly available information on supplier costs. Facilities and Administrative (F&A) costs were projected to rise over the first half of the project and then fall off as project completion passes the halfway point. The mission-specific constraints outlined in the mission task documents were integral to the budget development, particularly given the specialized equipment and protocols required for research in the permanently shaded lunar regions.

1.10.3. Personnel Budget

The following chart gives a breakdown of the number of people on each subteam over the phases of the mission. During the Mission lifecycle, the first year of Phase C requires less employment as the project ramps up, and then during years two, three, and four require more employment during Phases C and D. Then, as most of the building is completed and the project becomes more focused on launch, there is a slight decrease in the number of personnel, seen in years five and six and during Phase E and F, respectively.

Broken down by role, the science, engineering, and technician personnel follows this trend of increasing during years 2-4 before a decrease again. Administration and management also follow this trend, though the number of employees in each role is far smaller, with a maximum of four personnel being on the team during any part of the mission.

Figure 11. Number of Personnel

	Additional Information					
	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F
# People on Team	FY 1	FY 2	FY 3	FY 4	FY 5	FY 6
Science Personnel:	6	14	14	14	8	8
Engineering Personnel:	8	18	18	18	10	10
Technicians:	4	9	9	9	5	5
Administration Personnel:	2	4	4	4	2	2
Management Personnel:	2	4	4	4	2	2

The budgets for the personnel follow the trends described in Figure 9. Engineering has the highest total cost, followed in order by science, technicians, project management, and administration, due to the number of employees in each role and the cost to employ that role. Salaries total to $18,467,320. Subsequently for each role, there is an ERE that totals to $5,154,229 and a personal margin that totals to $18,000. This puts the total personnel budget at $25,152,201.

Figure 12. Personnel Budgets

Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Cumulative Total
PERSONNEL							
Science Personnel	$ 480,000	$ 1,149,120	$ 1,178,240	$ 1,207,360	$ 706,560	$ 723,200	$ 5,444,480
Engineering Personnel	$ 640,000	$ 1,477,440	$ 1,514,880	$ 1,552,320	$ 883,200	$ 904,000	$ 6,971,840
Technicians	$ 240,000	$ 554,040	$ 568,080	$ 582,120	$ 331,200	$ 339,000	$ 2,614,440
Administration Personnel	$ 120,000	$ 246,240	$ 252,480	$ 258,720	$ 132,480	$ 135,600	$ 1,145,520
Project Management	$ 240,000	$ 492,480	$ 504,960	$ 517,440	$ 264,960	$ 271,200	$ 2,291,040
Total Salaries	$ 1,720,000	$ 3,919,320	$ 4,018,640	$ 4,117,960	$ 2,318,400	$ 2,373,000	**$ 18,467,320**
Total ERE	$ 480,052	$ 1,093,882	$ 1,121,602	$ 1,149,323	$ 647,065	$ 662,304	**$ 5,154,229**
Personnel Margin	$ 3,000	$ 3,000	$ 3,000	$ 3,000	$ 3,000	$ 3,000	**$ 18,000**
TOTAL PERSONNEL	$ 2,203,052	$ 5,146,623	$ 5,410,691	$ 5,681,365	$ 3,277,186	$ 3,433,284	$ 25,152,201

1.10.4. Travel Budget

The travel budget can be broken down into the total amount spent on flights, hotels, transportation, per diem cost, and a margin for the various travel parts. The per diem costs are the highest of any of the categories, which are based on the rates for Cape Canaveral found on the GSA website. The GSA website allows for estimates to be calculated for travel.

The travel costs make up a small percentage of the total budget, however, the travel budget is a necessary component of all phases, especially the Phase E which is important to the launch of the rover. Traveling to locations is important for all phases, but especially for Phase E, since more staff will be needed on site, where they will all therefore need to be lodged.

Figure 13. Travel Budget Breakdown

Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Cumulative Total
TRAVEL							
Total Flights Cost	$ 8,000	$ 8,000	$ 8,000	$ 8,000	$ 14,800	$ 8,000	**$ 54,800**
Total Hotel Cost	$ 9,472	$ 10,212	$ 11,988	$ 11,988	$ 22,940	$ 7,992	**$ 74,592**
Total Transportation Cost	$ 1,350	$ 450	$ 600	$ 600	$ 2,550	$ 1,050	**$ 6,600**
Total Per Diem Cost	$ 13,040	$ 16,056	$ 18,501	$ 18,501	$ 28,281	$ 11,003	**$ 105,382**
Travel Margin	$ 1,000	$ 1,000	$ 1,000	$ 1,000	$ 1,000	$ 1,000	**$ 6,000**
Total Travel Costs	$ 32,862	$ 36,647	$ 42,174	$ 43,216	$ 76,806	$ 32,821	**$ 264,525**

1.10.5. Outreach Budget

The total outreach budget will cost $2,232,787. This will amount to approximately 1% of the total budget. The categories of budgets will be divided into the main sections of digital media outreach, in person outreach, and news outreach.

Figure 14. Outreach Budget Breakdown

OUTREACH							
Total Outreach Materials	$ 26,000	$ 56,000	$ 84,000	$ 110,000	$ 297,000	$ 26,000	$ 599,000
Total Outreach Venue Costs	$ 1,000	$ 35,000	$ 50,000	$ 70,000	$ 190,000	$ 2,500	$ 348,500
Total Outreach Travel Costs	$ 6,000	$ 20,000	$ 30,000	$ 40,000	$ 120,000	$ 10,000	$ 226,000
Total Outreach Services Costs	$ 16,000	$ 30,000	$ 50,000	$ 60,000	$ 190,000	$ 15,000	$ 361,000
Total Outreach Personnel Costs	$ 16,000	$ 30,000	$ 50,000	$ 60,000	$ 190,000	$ 15,000	$ 361,000
Outreach Margin	$ 6,000	$ 10,000	$ 25,000	$ 30,000	$ 90,000	$ 6,000	$ 167,000
Total Outreach Costs	$ 71,000	$ 185,706	$ 304,028	$ 398,860	$ 1,189,008	$ 84,185	$ 2,232,787

News Outreach

This includes costs associated with outreach associated with reaching out to traditional and nontraditional media outlets. This includes compensation for creative works, especially podcasts and blogs commissioned to report on events. This will be allocated a budget of $25,000 as this is expected to take up a small section of the outreach efforts. There will also be $5,000 of margin for this outreach budget category to ensure necessary expenditures are met.

Digital Media Outreach

This includes costs associated with social media, digital content creation, and creating lesson plans and citizen science projects. The total budget for this section will be allocated $520,000.

Social media efforts will be allocated $75,000 for the purpose of creating and distributing posts on social media platforms. This includes the creation of graphics and writing the body of the post as well as moderating the posts after publishing.

$300,000 will be allocated for video production with $100,000 of that being allocated to short form video production and $200,000 being allocated for long form video production. Video production funds will go towards writing, creating sets, recording, editing, and publishing these videos.

$50,000 will be allocated to creating materials to provide materials and resources to parents and teachers such as K-12 lessons with activities and experiments, coloring pages, and home experiments and activities

$25,000 will be allocated to maintaining and moderating citizen science projects related to the mission. And $70,000 will be allocated for margin in this category.

In person outreach

In person outreach includes expenses related to in person demonstrations as well as internship expenses. $520,000 will be allocated to these expenses.

$200,000 will be allocated towards in person outreach demonstrations such as creating models of the rover, posters, stickers, flyers, as well as any other needed materials for demonstration purposes and finally, $250,000 will be allocated towards internships.

1.10.6. Direct Costs

The direct manufacturing costs for each subteam are based on the components and subsystems that the subteam needs. The size, weight, and materials used are all important indicators of how much a component will cost. This section will review the direct costs associated with each subteam and where these costs are coming from within each subteam.

Figure 15. Direct Cost Breakdown

Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
DIRECT COSTS							
Mechanical Subsystem	$ 1,398,024	$ 2,153,712	$ 2,115,927	$ 1,586,946	$ 226,706	$ 75,568	$ 7,556,883
Power Subsystem	$ 1,900,300	$ 6,105,189	$ 5,998,080	$ 4,498,560	$ 642,651	$ 214,217	$ 19,358,997
Thermal Control Subsystem	$ 471,401	$ 726,212	$ 713,471	$ 535,103	$ 76,443	$ 25,481	$ 2,548,111
Comms & Data Handling Subsystem	$ 136,054	$ 209,597	$ 205,920	$ 154,440	$ 22,062	$ 7,354	$ 735,427
Science Instrumentation	$ 7,141,488	$ 11,001,752	$ 10,808,739	$ 8,106,554	$ 1,158,079	$ 386,026	$ 38,602,638
Spacecraft Cost Margin	$ 2,100,000	$ 3,000,000	$ 3,000,000	$ 1,600,000	$ 300,000	$ 300,000	$ 10,300,000
Total Spacecraft Direct Costs	$ 13,147,267	$ 23,799,570	$ 24,029,928	$ 17,767,168	$ 2,678,239	$ 1,139,770	$ 82,561,942
Manufacturing Facility Cost	$ 5,026,394	$ 7,180,562	$ 7,180,562	$ 3,829,633	$ 718,056	$ -	$ 23,935,208
Test Facility Cost	$ 4,179,000	$ 5,970,000	$ 5,970,000	$ 3,184,000	$ 597,000	$ -	$ 19,900,000
Facility Cost Margin	$ 2,100,000	$ 3,000,000	$ 3,000,000	$ 1,600,000	$ 300,000	$ 300,000	$ 10,300,000
Total Facilities Costs	$ 11,305,394	$ 16,570,477	$ 16,990,392	$ 9,285,497	$ 1,783,022	$ 339,000	$ 56,273,781
Total Direct Costs	$ 24,452,661	$ 40,370,047	$ 41,020,320	$ 27,052,665	$ 4,461,261	$ 1,478,770	$ 138,835,723
Total MTDC	$ 13,147,267	$ 23,799,570	$ 24,029,928	$ 17,767,168	$ 2,678,239	$ 1,139,770	$ 82,561,942

Mechanical

The direct costs associated with the mechanical subsystem is estimated to be a total of $44,000,000 including all components (wheels, suspension, chassis, drill), materials, testing and personnel. This section will go over all direct costs and quantify each amount.

The chassis and suspension will be constructed in-house of aluminum 6061, a widely used material in different space applications. This material will provide all the necessary properties to perform effectively. The size of this component will be the largest of the three main components with dimensions 1 x .5 x .25 m. The cost of this material can be purchased in bulk sheets or solid blocks and tubing for machining at ~5 dollars per pound. Manufacturing costs and fabrication of the chassis and suspension are estimated to be approximately $28,000,000. This includes manufacturing, facility costs, materials, and personnel services.

The wheels are the final component to be considered will also be constructed through a NASA facility, with each wheel being constructed from a single block of aluminum and the estimated cost for each wheel is estimated to be ~166,000 dollars each or ~1 million dollars total for all 6 wheels, this would also include manufacturing, facility costs, and materials.
The TRIDENT drill is the final component of the mechanical system and can be contracted through a third party company Honeybee Robotics, this product has a mass of 20 kg and draws 5 W of power, this product is estimated to cost $ ~15,000,000 for manufacturing and testing. All of these costs were derived using the NICM formula.

Power

The direct costs of the power subsystem involve the parts of the system: the solar panel, battery pack, and combined solar charge converter / power supply, As well as the personnel required to assemble the system. In this section of the direct costs. the costs of each component of the subsystem and the cost to construct and assemble the subsystem.

Firstly, the largest direct cost of the system is the ZTJ solar panel which will cost approximately 1.5 million dollars. This was found by looking at previous missions that have used the ZTJ solar cell and also via indirect contact with the company. After the solar cell, the next most expensive item of the system is the battery pack; this is estimated at a total of 400000 dollars for 7 batteries as of now but may increase in cost and number because of expanding scope. After this the final piece of the power subsystem is the power supply; this component performs several roles such as the solar charge controller and converter from low voltage from the solar panels to high voltage that the dedicated portion of the PSU uses, however, it will likely cost the least with an estimated cost of 100000 dollars given by previous missions using the STARBUCK-MINI such as the Intuitive Machines 2024 lunar mission. The estimated total for this portion of the dedicated costs comes to around 2 million dollars and makes up the larger portion of the dedicated costs between components and personnel costs.

Next, the cost of the personnel to assemble this system was calculated by finding a rough estimate of a manufacturing and development timeline based off of other previous missions like NASA’s Perseverance rover through public updates. After finding this timeline which came out to two years between finishing system design and the end of manufacturing, the personnel for the subteam were selected and came out to two technicians at 60000 dollars a year, 2 engineers at 80000 dollars a year, and 1 project manager at 120000 dollars a year. The personnel costs for this stands at 400000 dollars a year for two years.

After calculating all of the costs, the cost of the power subsystem should come out to 2.8 million dollars for the components needed to manufacture the system, and the personnel needed to put the subsystem together.

Communications and Data Handling

The single-board computer, the BAE Systems RAD5545 will be supplied through a contractor. The estimated total cost is $1,900,000.00. The estimated total manufacturing cost is $1,400,000.00, and the estimated total testing facility cost is $500,000.00.
The SpaceWire data bus's supplier is STAR-Dundee. The mass/cost for the item is variable depending on the length of cables used, thus the maximum length of 10 meters was picked in order to have the safest margin. The estimated total cost is $2,200,000.00. The estimated total manufacturing cost is $1,700,000.00, and the estimated total testing facility cost is $500,000.00.
The antenna (transmitter+receiver), the BAE Systems Ka-band multi-orbit LEO optimized antenna, will be supplied through ViaSat. The estimated total cost will be $8,300,000.00. The estimated total manufacturing cost is $6,400,000.00, and the estimated total testing facility cost is $1,900,000.00.
The 2GB of radiation-hardened flash storage will be supplied by Philson USA. The estimated total cost will be $500,000.00. The estimated total manufacturing cost is $400,000.00, and the estimated total testing facility cost is $100,000.00.
The total subsystem cost is $12,900,000.00, with $9,900,000.00 coming from manufacturing and $3,000,000.00 coming from t
All of these costs were derived using the NICM formula for Electronics and Software and the MCCET.

Thermal

The heater, Minco Polyimide Thermofoil HK Series, will be supplied by their contractor. The estimated total cost is $500,000. Including manufacturing and labor cost can be estimated up to $300,000 and testing facility cost can be estimated $200,000.

The radiator, Redwire Space: Q-Rad will be manufactured by the main supplier. The estimated total cost for the radiator is $500,000. The estimated total manufacturing and labor cost will be $350,000 and the total testing facility cost can range from $100,000-250,000. This equation is taken into account with the four main factors of Total Cost=Material Cost+Labor Cost+Overhead+Testing and QA Cost.

The thermal switch Diaphragm Thin Plate will be manufactured and tested by Sierra Space. The estimated total cost for the thermal switch is $100,000. The estimated manufacturing and labor cost will be $80,000 and the testing cost will be $40,000. This equation is taken into account with factors of the product material, labor and facility testing within Sierra Space.

The sprayable thermal paint, Socomore A276 Polyurethane will be purchased from Socomore. One unit of a bucket of paint will cost $179.73. Labor and testing costs could be estimated to be up to $3,000-5,000.

Science

The two science instruments that make up the direct costs will be contracted through the same company, Astrobotic. The total direct costs for the following instruments are approximately $17,200,000.

The first instrument is the Neutron Spectrometer system (NSS), a COTS part to be developed through the selected contractor. This part has a mass of 1.6 kg and draws 1.5 W of power to operate. The total for this instrument is estimated at 3,700,000. The costs associated with this part would also require testing which can be estimated to be up to 50K.

The second instrument is the Neutron Infrared Volatile Spectrometer system (NIRVSS). This part will also be contracted by Astrobotic and has a mass of 3.6 kg and draws 30 W of power. The total for this instrument is estimated at $13,400,000. This would include the manufacturing and development through the mentioned contractor. Testing for this part can add up to an additional 50K. All of these costs were derived using the NICM formula.

1.11. Scope Management

1.11.1. Change Control Management

Our team's process for change control is to, first and foremost, communicate these changes in a way that everyone can be aware of. These methods can include email, personal messages, and verbal communication through daily meetings. If there are any updates to a change, then these updates will be brought up as soon as possible. Any miscommunications must be avoided as they will hinder the team in the long run.

Our plans for implementing customer-requested changes are as follows: After receiving an RFA or ADV, these changes will be tracked in a spreadsheet to track document changes. After an update, a document will gain a distinguisher marking, such as A, B, C, etc. These will be incremented through the alphabet when a change has occurred. These changes will be communicated throughout the team via the spreadsheet tracking the different versions of the documents and through communication channels throughout the team.

For internal-requested changes, our plans for implementing them are as follows: Feedback will always be encouraged from all team members. Any feedback necessitating filling out an RFA or ADV will be brought to the mission administration for review. Upon review, an RFA or ADV will be filled out and these changes will be tracked in the spreadsheet mentioned above. The process will continue as described in the previous paragraph.

1.11.2. Scope Control Management

Several methods will be implemented if the team must downslope to correct overrun costs or schedules. First, the team will conduct subsystem evaluation meetings, similar to the previous process, to reduce the mission budget from $225 million to $175 million. These meetings will involve key stakeholders to assess the importance of specific components and alternative approaches.

Manufacturing and procurement methods will be reviewed to identify cost-saving opportunities while keeping quality high. This might include switching to more cost-effective suppliers, changing manufacturing methods, changing materials, or changing physical characteristics.

Non-essential activities such as travel, outreach, and logistics will be evaluated for budget reductions, with funds redirected to more critical mission components to ensure the mission’s success. As a last resort, specific science instruments may be removed to prioritize instruments with the highest scientific value.

Conversely, if the team has an additional margin for budget, funds will be used to enhance existing capabilities and subsystem performance. Additional budget may be used to add new science instruments, expand outreach and public engagement activities, improve redundancy and reliability, and enhance data analysis and processing capabilities.

The approach to scope control will vary depending on the mission life cycle phase. In the early stages, thorough planning and design reviews will identify potential risks and cost-saving opportunities, while mid-mission phases will focus on monitoring progress and implementing corrective actions. In the late phases, priority will be given to testing and validation to ensure mission readiness and address any last-minute issues or changes.

1.12. Outreach Plan

The purpose of the outreach directive of the mission is to inform the taxpayer of the mission as well as the science obtained from the mission.

News Outreach Efforts

The first aspect of outreach to consider is to ensure positive publicity in traditional and nontraditional news organizations when approaching key mission points or discoveries such as launch or critical science discoveries. These efforts will target mainly adults and have two categories of audiences which are general audiences and technically inclined audience members with interest in space exploration. Traditional news will target a mainstream, general audience. In this context, traditional media includes print, broadcast, radio, etc. Reaching out to national and local media distributors and journalists will ensure people of a range of backgrounds are aware of the mission.

Non-traditional media includes space themed news podcasts, blogs, etc. These outlets are geared towards people already interested in space news and information. By spreading information through these avenues, the team will engage with communities that are passionate about the mission who are likely to share technological successes as well as scientific discoveries with friends and family.

Digital Media Outreach Efforts

In addition, including digital media efforts in outreach will be critical to interacting with and informing a broad audience. Managing social media pages on mainstream sites and posting about mission updates as well as science discoveries made in a timely fashion enhances the ability to communicate with the taxpayer and allows for engagement and involvement.

This effort would target social media users, which are aged 13 or older. These people may or may not have a technical or scientific background so communicating in an engaging and easy to understand manner will be crucial. Posts will include mission updates and will usually be accompanied by a graphic or video relating to the topic being discussed in the body of the post. Platforms utilized can include Facebook, X, Instagram. Others may be added to this list as necessitated or decided by the team. YouTube will be discussed further in this section.

It is important to utilize existing NASA social media presence in the outreach efforts as applicable, such as the official NASA page, NASA Center pages, and any other page deemed to be appropriate.

Video presence will be important for outreach purposes of the mission. Videos can be addressed to adults as well as K-12 students. There are two types of videos under consideration. The first, short form videos, in this context a video less than 10 minutes. The second is long form videos, in this context a video with a length that is greater than 10 minutes.

Short form videos will be focused on engineering or science topics related to the mission. They will be designed to be approachable by children or anyone with rudimentary technical knowledge. The aim of these videos will be to increase general scientific literacy, while still focusing on aspects specifically related to the mission. These videos will make up a series of videos. These videos will be accessible on YouTube, the NASA website, as well as on NASA+. There is also the possibility of creating short videos related to special events for the purpose of increasing visibility such as videos relating to launch or landing on the moon.

Long form videos will be focused on the design and operation of the rover as well as scientific discoveries derived from received data. The aim of these videos will be to inform the general public of the work being done within the team as well as inspire the next generation of scientists and engineers. These videos will result in the creation of a documentary series available on YouTube, the NASA website, as well as NASA+. This series will be released at or some time after the rover arrives on the lunar surface.

Creating classroom aids for teachers as well as parents will be critical to reaching K-12 students. A curation of classroom lessons and activities related to the mission will be made available to teachers for the purpose of distributing and instructing students about science and engineering principles related to the mission. These lessons and activities will include powerpoint presentations as well as interactive crafts or experiments to perform. There will also be a science breakdown instructing teachers on the basic science principles of the phenomena described in the lesson, experiment, or craft provided. While each state may have a different science curriculum for students, keeping education standards in the creation of these resources will be imperative to adding as much value to teachers and students as possible.

At-home activities such as coloring books, science experiments, and crafts will make the mission accessible to parents and children. These resources will include instructions, explanations about the relevant science or engineering topics related to the mission, as well as further resources for inquisitive children or parents interested in learning more. The coloring pages will feature the rover and its various subsystems with brief descriptions of the system(s) depicted as well as highlighting the science objectives of the mission. The science experiments and crafts will demonstrate relevant science and engineering principles related to the rover and the science objectives of the mission. This will provide children and parents with interactive and inexpensive means of engaging with the mission and the science objectives.

Citizen science will be an interesting and engaging way of enabling volunteers, particularly highschool and college students, to interact directly with the mission. Presenting data from the rover in a way that ordinary people can contribute to the science will provide meaningful experiences for regular people to engage with research directly related to the mission. Zooniverse would be a great resource to host this citizen science opportunity.

In Person Outreach Efforts

Having in person outreach efforts, specifically near key mission milestones, such as launch or landing on the lunar surface as these in person efforts will be most impactful adjacent to events such as launch or landing on the lunar surface. Large events will ideally feature small scale models of the rover, informational posters, a video presentation, small samples of lunar regolith for view, one or two interactive demonstrations, and items to distribute such as stickers or flyers. Small events can feature as little as informational posters, one or two interactive demonstrations, and items to distribute such as stickers or flyers.
Large events will be performed directly by the team and will take place near key mission milestones. Other demonstrations will be facilitated by the contracted partners as well as volunteers from the community. Large events will take place in public places such as NASA centers or planetariums. Smaller events can take place at STEM community nights for universities or schools as well as other events as deemed appropriate by the team.

The team will also offer internships and opportunities for college or highschool students to participate in the work being performed on the mission. This will engage a variety of different individuals and develop the incoming STEM workforce. Engaging in this way will provide hands-on opportunities for interested emerging workers to engage with their specific field of study as well as introduce the work environment in a technical field and specifically at NASA.

Partner Outreach

The team will utilize mission partners in outreach efforts. Ensuring that partners share this team's outreach materials as well as enabling them to share resources related to their own contributions to the mission will increase the efficacy of the mission's outreach efforts as well as reach individuals specific to regions where mission partners are located.

1.13. Conclusion

The Ice Hunter's MDR outlines the team's plans to procure the necessary parts for the rover. This includes figuring out the suppliers or procedures (including NASA itself) for all parts of the rover. Furthermore, the MDR outlines the costs associated with all parts of the rover, including the costs of all the rover subteams. and subsystems within these subteams. Furthermore, budgets were created for the personnel needed, along with costs to support the mission and its staff, such as a travel budget and an outreach budget. To further prove viability, the team defined its structure of change control and scope control, along generally describing its approach to the team and its structure.

The Ice Hunters are working towards updating all sections with complete information, as some sections lack the necessary

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the team used ChatGPT in order to estimate lead times for the CDH components. After using this tool, the team reviewed the content and edited it. On top of this, ChatGPT was used to create general citations for the Power Subsystem which were then checked and corrected by hand. ChatGPT was also used to convert a table of positions required for each phase into textual descriptions which was then edited to describe the roles of each position during each phase.

Appendix

CDH Direct Cost Derivation:

Electrical CER	Software CER
1516 * 0.24^0.74 = 527.297	236 * 0.24^0.69 = 88.157
1516 * 2.45^0.74 = 2942.27	236 * 2.45^0.69 = 437.96
1516 * 0.3^0.74 = 621.96	236 * 0.3^0.69 = 102.83
1516 * 0.03^0.74 = 113.18	236 * 0.03^0.69 = 20.99

This is the derivation for the CER numbers for CDH. These numbers were inputted into the MCCET to determine costs for all of the components.